

Circulatory Systems

Subject & topics: Biology | Physiology | Breathing & Circulation **Stage & difficulty:** A Level P2

Unicellular organisms rely on diffusion, osmosis, and active transport to take in what they need from their environment and expel waste products to their environment. However, most multicellular organisms cannot solely rely on these processes. They have circulatory systems that transport substances throughout the organism.

Part A

The need for circulatory systems

Why do larger organisms need a circulatory system? Select all that apply.

- ☐ Larger organisms have a **higher** surface area to volume (SA : V) ratio, which means that the efficiency of diffusion is lower.
- ☐ Larger organisms have a **lower** surface area to volume (SA : V) ratio, which means that the efficiency of diffusion is lower.
- ☐ Oxygen and carbon dioxide are exchanged by active transport in multicellular organisms, not by diffusion.
- ☐ Diffusion over large distances (e.g. across tissues and organs) would be too slow to provide oxygen/nutrients quickly enough for respiration.
- ☐ Multicellular organisms need oxygen for respiration, whereas unicellular organisms do not.
- ☐ Unicellular organisms live in a fluid environment, whereas all multicellular organisms live on land.

Part B

Open vs closed circulatory systems

Which of the following are true of an **open** circulatory system? Select all that apply.

- ☐ the transport medium is called blood
- ☐ the transport medium is called haemolymph
- ☐ the transport medium is under **lower** pressure than in a closed circulatory system
- ☐ the transport medium is under **higher** pressure than in a closed circulatory system
- ☐ the transport medium is pumped from the heart through arteries, then capillaries, then veins, and then flows back into the heart
- ☐ the transport medium is pumped from the heart out into the main body cavity of the animal, and then flows back into the heart
- ☐ the transport medium comes into direct contact with the tissues it delivers nutrients to
- ☐ the transport medium does **not** come into direct contact with the tissues it delivers nutrients to
- ☐ found in insects
- ☐ found in vertebrates and annelids

Which of the following are true of a **closed** circulatory system? Select all that apply.

- ☐ the transport medium is called blood
- ☐ the transport medium is called haemolymph
- ☐ the transport medium is under **lower** pressure than in an open circulatory system
- ☐ the transport medium is under **higher** pressure than in an open circulatory system
- ☐ the transport medium is pumped from the heart through arteries, then capillaries, then veins, and then flows back into the heart
- ☐ the transport medium is pumped from the heart out into the main body cavity of the animal, and then flows back into the heart
- ☐ the transport medium comes into direct contact with the tissues it delivers nutrients to
- ☐ the transport medium does **not** come into direct contact with the tissues it delivers nutrients to
- ☐ found in insects
- ☐ found in vertebrates and annelids

Single vs double circulatory systems

Which of the following is true of a **single** circulatory system? Select all that apply.

- ☐ blood is pumped from the heart to the main respiratory organs (e.g. gills/lungs) and then returns back to the heart before being pumped around the rest of the body
- ☐ blood is pumped from the heart to the main respiratory organs (e.g. gills/lungs) and then flows to the rest of the body before returning to the heart
- ☐ blood flows throughout the body at a **lower** pressure and **slower** speed than in a double circulatory system
- ☐ blood flows throughout the body at a **higher** pressure and **faster** speed than in a double circulatory system
- ☐ found in reptiles, birds, and mammals
- ☐ found in fish

Which of the following is true of a **double** circulatory system? Select all that apply.

- ☐ blood is pumped from the heart to the main respiratory organs (e.g. gills/lungs) and then returns back to the heart before being pumped around the rest of the body
- ☐ blood is pumped from the heart to the main respiratory organs (e.g. gills/lungs) and then flows to the rest of the body before returning to the heart
- ☐ blood flows throughout the body at a **lower** pressure and **slower** speed than in a single circulatory system
- ☐ blood flows throughout the body at a **higher** pressure and **faster** speed than in a single circulatory system
- ☐ found in reptiles, birds, and mammals
- ☐ found in fish

Part D

Organisms

Match the organism to the characteristic in the table below.

Characteristic	Organism
contains haemolymph	
blood flows in and out of the heart twice for every time it reaches a particular tissue	
blood flows in and out of the heart once for every time it reaches a particular tissue	
does not contain a circulatory system	

Items:

- grasshopper
- goldfish
- pigeon
- Amoeba*

Human Heart Anatomy

Subject & topics: Biology | Physiology | Breathing & Circulation Stage & difficulty: A Level P2

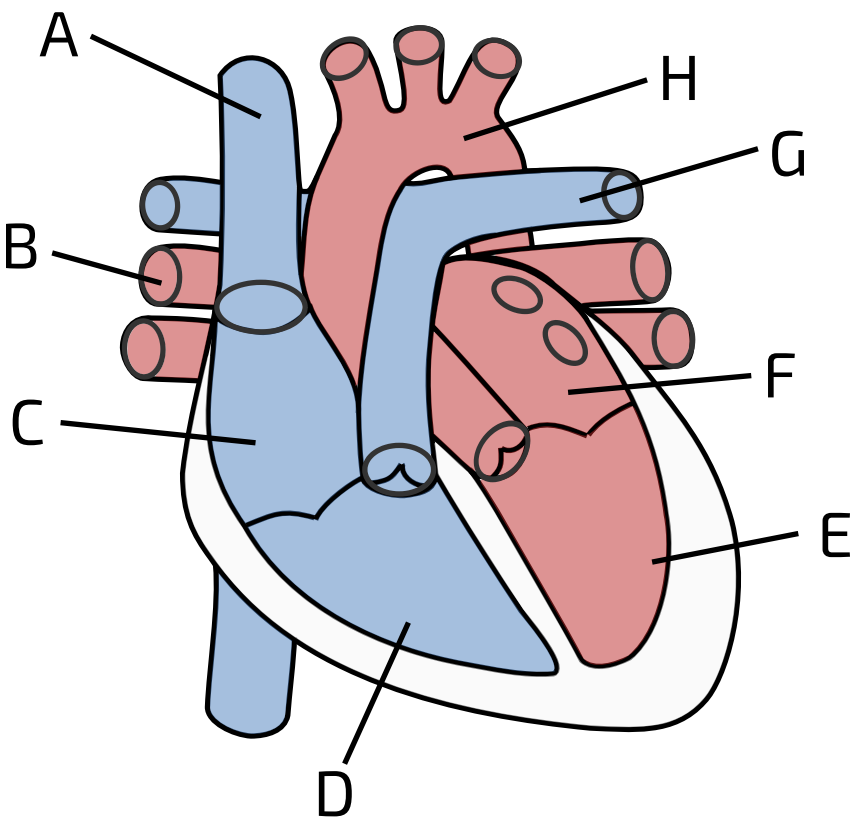


Figure 1: Diagram of the inside of the human heart. Regions are colour-coded by oxygen concentration (blue = low, red = high). The heart is shown in cross-section from the front.

Part A

Label the heart

Letter	Structure
A	
B	
C	
D	
E	
F	
G	
H	

- Items:
- vena cava

a pulmonary artery

aorta

right ventricle

left ventricle

left atrium

a pulmonary vein

right atrium

Part B

Direction of blood flow

Drag the letters on the left into the correct order on the right to show the direction of blood flow through the heart. Begin (at the top) with the letter that represents the vena cava.

Available items

A

B

C

D

E

F

G

H

Part C

Heart valves

What is the general name for the valves that separate **C** from **D** and **F** from **E**?

What is the general name for the valves that separate **D** from **G** and **E** from **H**?

The Cardiac Cycle

Subject & topics: Biology | Physiology | Breathing & Circulation **Stage & difficulty:** A Level P2

The cardiac cycle is the sequence of events that a heart undergoes from one heartbeat to the next.

Part A

Blood flow

Drag the items below into the correct order on the right to show the path that blood flows through the human body.

Begin (at the top) with "vena cava". You should use all of the available items.

Available items

- vena cava
- aorta
- capillaries in the lungs
- capillaries in the tissues/organs of the rest of the body
- left atrium
- left ventricle
- pulmonary arteries
- pulmonary veins
- right atrium
- right ventricle
- systemic arteries
- systemic veins

Part B

Systole & Diastole

There are two main phases of the cardiac cycle: systole (contraction) and diastole (relaxation). Fill in the blanks below to describe the cardiac cycle for an organism with a double circulatory system.

Systole

First, the atria contract (atrial systole) to pump blood into the ventricles. This contraction is caused by electrical excitation spreading from the .

Next, the ventricles contract (ventricular systole), causing the atrioventricular valves to and the semi-lunar valves to , and forcing blood out of the heart into the aorta and pulmonary artery.

This contraction is caused by electrical excitation in the that spreads through the bundle of His (a bundle of conducting tissue made up of modified muscle fibres called Purkyne fibres).

The semi-lunar valves then , preventing backflow of blood into the ventricles.

Items:

- close
- atrio-ventricular node (AVN)
- open
- sino-atrial node (SAN)

Diastole

During diastole, the atria and the ventricles relax together (though the atria were already starting to relax during). Blood flows from the vena cava and pulmonary veins into the atria because of this relaxation. This causes the valves to , allowing blood to flow from the atria into the .

Items:

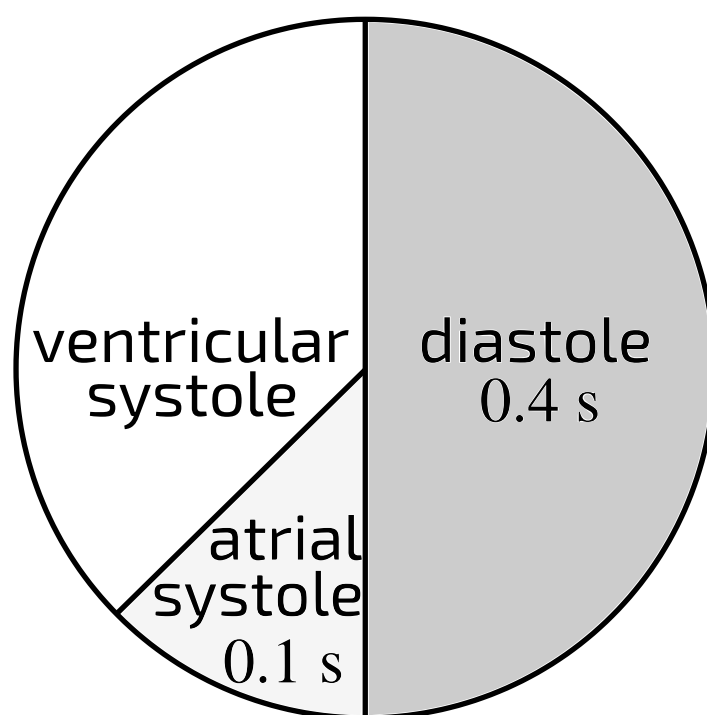
- ventricles
- aorta & pulmonary artery
- ventricular systole
- atrioventricular
- close
- open
- semilunar
- atrial systole

Part C

Heart rate

The chart below shows the stages of one cardiac cycle for an adult at rest.

Timings are shown for some of the stages.



Calculate the heart rate for this individual in bpm (beats per minute).

Created for isaacphysics.org by Lewis Thomson. Part C adapted with permission from OCR AS level Biology B June 2016 Foundations of biology Q4.

Question deck:

STEM SMART Biology Week 19 - Circulatory Systems 1

Heart Pressure Changes

Subject & topics: Biology | Physiology | Breathing & Circulation Stage & difficulty: A Level C2

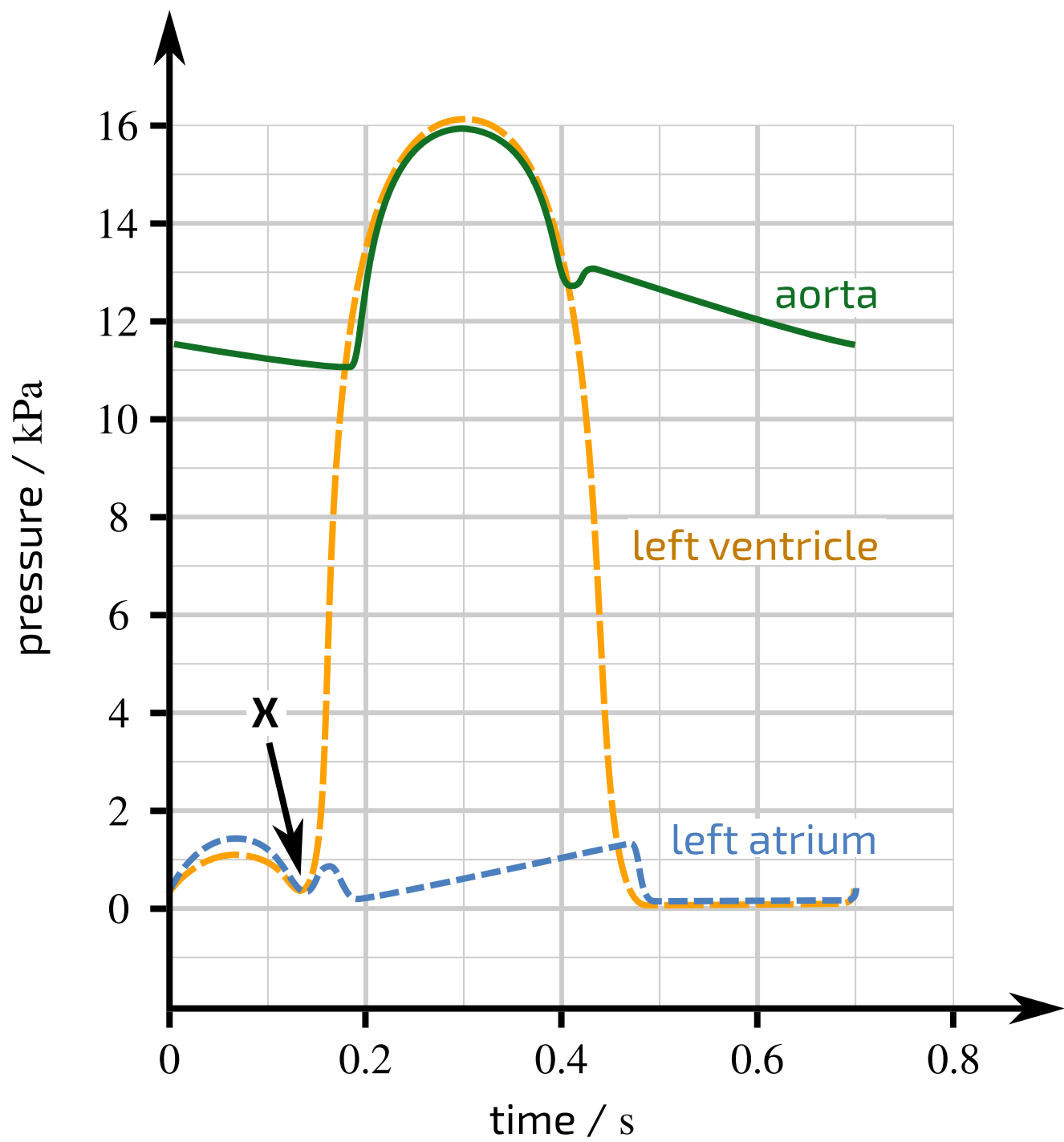


Figure 1: Pressure changes in the left atrium, left ventricle, and aorta during a single cardiac cycle.

Part A
0 s to point X

Which of the following statements are true of the period from 0 s to point **X** in **Figure 1**? Select all that apply.

- ☐ atrial systole is occurring
- ☐ atrial diastole is occurring
- ☐ pressure rises in the left atrium because it is contracting
- ☐ pressure rises in the left atrium because blood is moving into it from the pulmonary vein
- ☐ pressure rises in the left ventricle because it is contracting
- ☐ pressure rises in the left ventricle because blood is being pumped into it from the left atrium

Part B
Point X to 0.4 s

Which of the following statements are true of the period from point **X** to 0.4 s in **Figure 1**? Select all that apply.

- ☐ atrial systole is occurring
- ☐ ventricular systole is occurring
- ☐ pressure rises in the left ventricle because it is contracting
- ☐ pressure rises in the aorta because blood is being pumped into it from the left ventricle
- ☐ pressure rises in the left atrium because it is contracting
- ☐ pressure rises in the left atrium because blood is moving into it from the pulmonary vein

Part C

Heart rate

Calculate the heart rate of the individual shown in **Figure 1**.

Give your answer to 2 significant figures.

Part D

Aortic pressure

Calculate the percentage increase from minimum to maximum pressure in the aorta of the individual shown in **Figure 1**.

Give your answer to 2 significant figures.

Part E

Valves

Name the type of valve that closes at point **X** in **Figure 1**.

Adapted with permission from OCR A Level Biology A, June 2019, Biological processes, Question 16

Question deck:

STEM SMART Biology Week 19 - Circulatory Systems 1

Cardiac Output Calculations

Subject & topics: Biology | Physiology | Breathing & Circulation **Stage & difficulty:** A Level C2

Part A
Human daily cardiac output

The stroke volume of a particular person is 70 cm^3 , and their average heart rate is 75 bpm.

Calculate the daily cardiac output of this person.

Part B
Blue whale stroke volume

The average heart rate of a particular blue whale is 15 bpm. The cardiac output of this whale is $1200\text{ dm}^3\text{ min}^{-1}$.

What volume of blood is pumped out of the left ventricle during a single heart beat?

Part C
Hummingbird heart rate

The cardiac output of a particular hummingbird is 108 ml min^{-1} , and its stroke volume is $90\text{ }\mu\text{l}$.

Calculate the average heart rate of this hummingbird in beats per minute (bpm).

Part D

Another human's cardiac output

Immediately before ventricular systole, the combined volume of blood in both ventricles of a particular person is 280 ml.

Immediately after ventricular systole, the combined volume of blood in both ventricles is 100 ml.

Ventricular systole in this person takes 0.3 s, which represents 40% of their cardiac cycle duration.

Calculate the cardiac output of this person in $\text{dm}^3 \text{ min}^{-1}$.

Assume that both ventricles pump out equal volumes of blood during ventricular systole.

ECG Analysis

Subject & topics: Biology | Physiology | Breathing & Circulation Stage & difficulty: A Level C2

An electrocardiogram (ECG) shows the electrical activity of the heart. The graph below shows the changes in the volume of the ventricles, and the ECG tract that accompanies those changes, during two consecutive heart beats in an adult human.

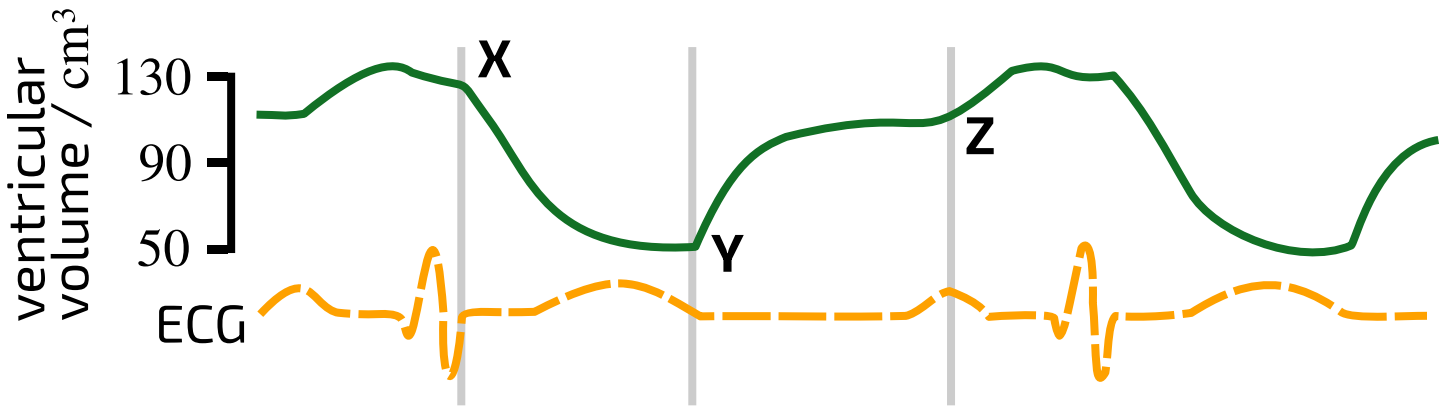


Figure 1: Volume of the ventricles over time and the accompanying ECG trace during two consecutive heart beats in an adult human.

Part A Blood movement

Fill in the blanks below to describe what is happening at points X, Y, and Z in **Figure 1**.

At **X** blood is being pumped into the

At **Y** valves between atria and ventricles are

At **Z** the chambers of the heart that are contracting are the

Items:

- atria
- aorta & pulmonary artery
- aorta & pulmonary vein
- pulmonary artery only
- pulmonary vein only
- ventricles
- open
- closed

Part B

Diastole & systole

Match the heart process to the point in **Figure 1** and the corresponding ECG wave(s) in the table below.

Diastole/systole	Figure 1 occurrence	Corresponding ECG wave(s)
Diastole		
Atrial systole		
Ventricular systole		

Items:

- between X and Y
- between Y and Z
- between Z and the point after Z that is equivalent to X
- P wave
- QRS complex
- T wave

Part C

Blood circulation

Based on **Figure 1**, how long would it take to pump all of the blood round the entire circulatory system?

Assume that

- the total blood volume is 5 litres
- the heart rate is 60 bpm (beats per minute)
- both ventricles pump out equal volumes of blood during ventricular systole

Give your answer to the nearest minute

Adapted with permission from NSAA 2020 Section 1 Q75